Magnetometer Calibration Using Levenberg-Marquardt Algorithm

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1 Introduction

A typical flight controller has several sensors for its attitude, heading, and position determination. A commonly found sensors are including 3-axis accelerometer, 3-axis gyroscope, 3-axis magnetometer, a pressure sensor, and GPS. From all those sensors, 3-axis magnetometer is one of the most critical sensors to determine drone's attitude and heading.

An ideal magnetometer measurement provides users with a constant magnitude in all direction, depending on its location on earth. When plotted, this measurement can be seen as a perfect sphere with origin at O (0,0,0). In reality, most magnetometer is uncalibrated and effected by its surrounding, making the measurement off from the center and distorted.

In general, there are two parameters contributing to an uncalibrated magnetometer measurement. The first one is **hard iron bias**, comes from a permanent magnetic filed and causes offsets in measurements. This is the largest contribution to an uncalibrated measurement and it comes from magnets or magnetized metal as well as high current wires. The second one is **soft iron distortions**, caused by paramagnetic materials, such as ferrous metals, around the magnetometer. This disturbance makes the measurement translated, shrunk, and possibly rotated.

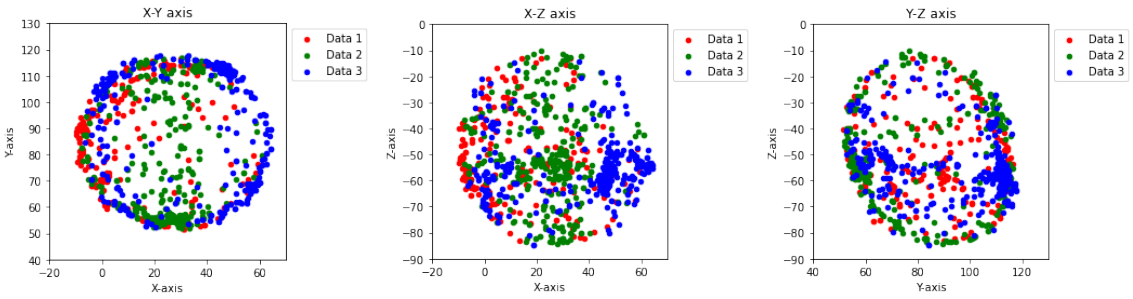

Figure 1: Uncalibrated Magnetometer Measurement

Figure 1 shows an uncalibrated magnetometer measurement from a low cost sensors in a 3-plane view. In all planes, one can see that the measurement is no centered at its origin $\boldsymbol{O}$. Plane X-Y shows that the measurement is distorted and formed an ellipse rather than a circle. On the X-Z plane, the measurement is only slightly distorted. Lastly, the elliptic distortion is also observable in the Y-Z plane, with and additional rotational effect.

2 Mathematical Model

As what has been identified in the previous section, a hard iron bias and a soft iron distortion contribute in a distorted and biased magnetometer measurement. Mathematically, one can model the hard iron bias as an offset (or bias) vector, while the soft iron distortion can be modelled as a 3x3 matrix. The equation of a calibrated measurement, considering both effect, can be seen in Equation 1.

$$\mathbf{h_{cal}} = K\left(\mathbf{h} - \mathbf{b}\right) \tag{1}$$

$$\begin{bmatrix} h_{x_{cal}} \\ h_{y_{cal}} \\ h_{z_{cal}} \end{bmatrix} = \begin{bmatrix} K_{dx} & K_{ox} & K_{oy} \\ K_{ox} & K_{dy} & K_{oz} \\ K_{oy} & K_{oz} & K_{dz} \end{bmatrix} \left(\begin{bmatrix} h_x \\ h_y \\ h_z \end{bmatrix} - \begin{bmatrix} b_x \\ b_y \\ b_z \end{bmatrix} \right) \tag{2}$$

Matrix K is the soft iron distortion matrix while $\mathbf{b}$ is the hard iron bias. To have the calibrated measurement, 6 parameters from the matrix and 3 parameters from the bias vector needs to be calculated. A non-linear least squared optimization will be used to identify those parameters is discussed in the next section.

3 Non-Linear Least Squares Optimization

Given a vector function $\mathbf{f} : \mathbf{R^n} \mapsto \mathbf{R^m}$ with $m \geq n$, one wants to minimize $||\mathbf{f(x)}||$, or equivalently find equation 3.

$$\mathbf{x}^* = \mathrm{argmin}_{\mathrm{x}} F(\mathbf{x}) \tag{3}$$

where

$$F(\mathbf{x}) = \frac{1}{2}\sum_{i=1}^{m}(\mathbf{f_i}(\mathbf{x}))^2 = \frac{1}{2}||\mathbf{f}(\mathbf{x})||^2 = \frac{1}{2}\mathbf{f}(\mathbf{x})^T\mathbf{f}(\mathbf{x}) \tag{4}$$

There are several methods to solve the least squares problem, one of it is Levenberg-Marquardt Algorithm (LM or LMA) which will be discussed later. The method will require a derivative of F. Given that $\mathbf{f}$ is has continous second partial derivatives, one can write the Taylor expansion as shown in equation 5.

$$\mathbf{f}(\mathbf{x} + \mathbf{h}) = \mathbf{f}(\mathbf{x}) + \mathbf{J}(\mathbf{x})\mathbf{h} + O(||h||^2) \tag{5}$$

where, $\mathbf{J}\epsilon\mathbf{R}^{mxn}$ is the Jacobian. This is a matrix containing the first partial derivatives of the function components, as shown in equation 6

$$(\mathbf{J}(\mathbf{x}))_{ij} = \frac{\partial f_i}{\partial x_j} \tag{6}$$

In an optimal condition, the gradient of the multi-variable function is zero at that certain point. It can be written mathematically as shown in equation 7 or 8.

$$\frac{1}{2}\nabla||\mathbf{f}(\mathbf{x}^*)||^2 = 0 \tag{7}$$

$$\mathbf{J}(\mathbf{x}^*)^T\mathbf{f}(\mathbf{x}^*) = 0 \tag{8}$$

However, solving a non linear equation is very difficult. In most cases, one cannot solve the exact value and only obtain the "good" solution. Also, since a heuristic algorithms is used, it is not guaranteed that one will always find the result. Nonetheless, this approach usually works well in practice.

4 Levenberg-Marquardt Algorithm

The Levenberg-Marquardt Algorithm is also known as the damped least squares (DLS) method. It can be used to solve a non-linear least squares problem especially a a least squares curve fitting. This approach is interpolates between the Gauss-Newton Algorithm and the Gradient Descent method.

At any point z, one can form an affine approximation of function f given that x is near z. The affine approximation can be written as shown in equation 9. Since it is an affine approximation, one can use a linear least squares to minimize $||\hat{f}(x;z)||^2$. One can iterate z until the optimal condition is obtained.

$$\hat{\mathbf{f}}(\mathbf{x};\mathbf{z}) = \mathbf{f}(\mathbf{z}) + \mathbf{J}(\mathbf{x})(\mathbf{x}-\mathbf{z}) \tag{9}$$

The Levenberg-Marquardt Algorithm uses an iterative approach to find for the optimum point. At any iteration k, one needs to form an affine approximation of f at $x^{(k)}$, as shown in equation 10. One needs to find $x^{(k+1)}$ to minimize equation 11 where $\lambda^{(k)}$ is positive. Note that $x^{(k+1)}$ should not be too far from $x^{(k)}$, otherwise the affine approximation will no longer hold.

$$\hat{\mathbf{f}}(\mathbf{x};\mathbf{x}^{(k)}) = \mathbf{f}(\mathbf{x}^{(k)}) + \mathbf{J}(\mathbf{x})(\mathbf{x}-\mathbf{x}^{(k)}) \tag{10}$$

$$||\hat{\mathbf{f}}(\mathbf{x};\mathbf{x}^{(k)})||^2 + \lambda^{(k)}||\mathbf{x}-\mathbf{x}^{(k)}||^2 \tag{11}$$

Substituting equation 10 to 11, one obtain equation 12.

$$||\mathbf{f}(\mathbf{x}^{(k)}) + \mathbf{J}(\mathbf{x})(\mathbf{x}-\mathbf{x}^{(k)})||^2 + \lambda^{(k)}||\mathbf{x}-\mathbf{x}^{(k)}||^2 \tag{12}$$

The solution is shown in equation 13. Note that the inverse always exists since $\lambda^{(k)}$ is always positive. The convergence is obtained when $J(x)^T f(x^{(k)}) = 0$, making $x^{(k+1)} = x^{(k)}$ in which the optimal condition holds. However, one needs to be careful since the optimal condition might be a local optima rather than a global optima.

$$\mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} - (\mathbf{J}(\mathbf{x})^T\mathbf{J}(\mathbf{x}) + \lambda^{(k)}\mathbf{I})^{-1}\mathbf{J}(\mathbf{x})^T\mathbf{f}(\mathbf{x}^{(k)}) \tag{13}$$

One important part of LMA is adjusting the damping factor $\lambda^{(k)}$ which contribute to the convergence of the solution. If $\lambda^{(k)}$ is too big, $\mathbf{x}^{(k+1)}$ will be too close to $x^{(k)}$ thus making the convergence time consuming. However, if $\lambda^{(k)}$ is too small, $\mathbf{x}^{(k+1)}$ may be far from $\mathbf{x}^{(k)}$, resulting in a poor affine approximation.

The damping factor λ has to be adjusted in each iteration. If the cost (or residual) at $k+1$ is less than the one at k, $||\mathbf{f}(\mathbf{x}^{(k+1)})||^2 < ||\mathbf{f}(\mathbf{x}^{(k)})||^2$, then accept the new x value and reduce the damping factor λ. Otherwise, increase the damping factor λ and do not update $\mathbf{x}$.

5 Sphere Fit

Hard iron bias is one of the parameter to be identified in magnetometer calibration. This value determine the offset of each measurement and therefore correct it to make each measurement centered at the origin O. Also, one has to determine the radius of the sphere to describe it. One way to do the calibration is by using sphere fit approach based on Levenberg-Marquardt Algorithm.

First of all, one needs to identify the equation for the sphere fit algorithm. For the cost function, one can determine the proximity between the measurement and the real value by comparing the calibrated value and local the magnetic field strength. Equation 14 shows the cost function used for sphere fit algorithm, where r is the local magnetic field strength.

$$f(\mathbf{x}) = r - ||\mathbf{h}_{cal}|| \tag{14}$$

where,

$$[\mathbf{x}] = \begin{bmatrix} r & b_x & b_y & b_z \end{bmatrix}^T \tag{15}$$

Substituting equation 2 to 14, one obtain equation 16. Note that for a sphere, the matrix K is an identity matrix since there is no distortion or rotation to the sphere. Thus, equation 16 can be written as equation 17. By expanding equation 17, one can rewrite it as shown in equation 18.

$$f(\mathbf{x}) = r - ||K\,(\mathbf{h} - \mathbf{b})\,|| \tag{16}$$

$$f(\mathbf{x}) = r - ||\mathbf{h} - \mathbf{b}|| \tag{17}$$

$$f(\mathbf{x}) = r - \sqrt{(h_x - b_x)^2 + (h_y - b_y)^2 + (h_z - b_z)^2} \tag{18}$$

As part of the LMA algorithm, the Jacobian has to be calculated. The Jacobian of the sphere fit is written in equation 19. Substituting equation 18 to 19, one can obtain equation 20 which describe the Jacobian of the sphere fit.

$$\mathbf{J}(\mathbf{x}) = \begin{bmatrix} \frac{\partial f(\mathbf{x})}{\partial r} & \frac{\partial f(\mathbf{x})}{\partial b_x} & \frac{\partial f(\mathbf{x})}{\partial b_y} & \frac{\partial f(\mathbf{x})}{\partial b_z} \end{bmatrix} \tag{19}$$

$$\mathbf{J}(\mathbf{x}) = \begin{bmatrix} 1 & \frac{(h_x - b_x)}{||\mathbf{h}-\mathbf{b}||} & \frac{(h_y - b_y)}{||\mathbf{h}-\mathbf{b}||} & \frac{(h_z - b_z)}{||\mathbf{h}-\mathbf{b}||} \end{bmatrix} \tag{20}$$

At this point, all the mathematical formulations are ready. One can continue to writing the program.

6 Ellipsoid Fit

Most magnetometer measurement will be affected by its surrounding and resulting a non-sphere measurement. To reduce this soft iron distortion, one needs to calibrate the magnetometer using ellipsoid fit approach.

The cost function for the ellipsoid fit is the same as the sphere fit, shown in equation 16. The matrix K accommodates the soft iron distortion and is a symmetric matrix. Thus, one can rewrite

equation 16 as equation 21. Note that the radius is a constant that can be pre-computed based on users location on earth.

$$f(\mathbf{x}) = r - || \begin{bmatrix} K_{dx} * (h_x - b_x) + K_{ox} * (h_y - b_y) + K_{oy} * (h_z - b_z) \\ K_{ox} * (h_x - b_x) + K_{dy} * (h_y - b_y) + K_{oz} * (h_z - b_z) \\ K_{oy} * (h_x - b_x) + K_{oz} * (h_y - b_y) + K_{dz} * (h_z - b_z) \end{bmatrix} || \tag{21}$$

where,

$$[\mathbf{x}] = \begin{bmatrix} b_x & b_y & b_z & K_{dx} & K_{dy} & K_{dz} & K_{ox} & K_{oy} & K_{oz} \end{bmatrix}^T \tag{22}$$

Let us call the vector elements as A, B, and C for practical purpose. Thus, one can write equation 21 as in 23 and 24.

$$f(\mathbf{x}) = r - || \begin{bmatrix} A \\ B \\ C \end{bmatrix} || \tag{23}$$

$$f(\mathbf{x}) = r - \sqrt{A^2 + B^2 + C^2} \tag{24}$$

As part of the LMA algorithm, the Jacobian matrix has to be calculated. The Jacobian of an ellipsoid fit is written in equation 25.

$$\mathbf{J}(\mathbf{x}) = \begin{bmatrix} \frac{\partial f(\mathbf{x})}{\partial b_x} & \frac{\partial f(\mathbf{x})}{\partial b_y} & \frac{\partial f(\mathbf{x})}{\partial b_z} & \frac{\partial f(\mathbf{x})}{\partial K_{dx}} & \frac{\partial f(\mathbf{x})}{\partial K_{dy}} & \frac{\partial f(\mathbf{x})}{\partial K_{dz}} & \frac{\partial f(\mathbf{x})}{\partial K_{ox}} & \frac{\partial f(\mathbf{x})}{\partial K_{oy}} & \frac{\partial f(\mathbf{x})}{\partial K_{oz}} \end{bmatrix}^T \tag{25}$$

where,

$$\frac{\partial f(\mathbf{x})}{\partial b_x} = \frac{AK_{dx} + BK_{ox} + CK_{oy}}{\sqrt{A^2 + B^2 + C^2}} \tag{26}$$

$$\frac{\partial f(\mathbf{x})}{\partial b_y} = \frac{AK_{ox} + BK_{dy} + CK_{oz}}{\sqrt{A^2 + B^2 + C^2}} \tag{27}$$

$$\frac{\partial f(\mathbf{x})}{\partial b_z} = \frac{AK_{oy} + BK_{oz} + CK_{dz}}{\sqrt{A^2 + B^2 + C^2}} \tag{28}$$

$$\frac{\partial f(\mathbf{x})}{\partial K_{dx}} = -1 * \frac{A(h_x - b_x)}{\sqrt{A^2 + B^2 + C^2}} \tag{29}$$

$$\frac{\partial f(\mathbf{x})}{\partial K_{dy}} = -1 * \frac{B(h_y - b_y)}{\sqrt{A^2 + B^2 + C^2}} \tag{30}$$

$$\frac{\partial f(\mathbf{x})}{\partial K_{dz}} = -1 * \frac{C(h_z - b_z)}{\sqrt{A^2 + B^2 + C^2}} \tag{31}$$

$$\frac{\partial f(\mathbf{x})}{\partial K_{ox}} = -1 * \frac{A(h_y - b_y) + B(h_x - b_x)}{\sqrt{A^2 + B^2 + C^2}} \tag{32}$$

$$\frac{\partial f(\mathbf{x})}{\partial K_{oy}} = -1 * \frac{A(h_z - b_z) + C(h_x - b_x)}{\sqrt{A^2 + B^2 + C^2}} \tag{33}$$

$$\frac{\partial f(\mathbf{x})}{\partial K_{oz}} = -1 * \frac{B(h_z - b_z) + C(h_y - b_y)}{\sqrt{A^2 + B^2 + C^2}} \tag{34}$$

7 Results

This section consists of two parts, which are Sphere Fit and Ellipsoid Fit result. The data used for the calibration are the same, only the calibration method is different. At the end of the section, fitness parameter and damping factor is compared between the two methods two check the performance. The fitness parameter is the mean squared errors (MSE) as shown in equation 35.

$$MSE = \frac{1}{N} \sum_{n=1}^{N} (f(x))^2 \tag{35}$$

The data used in this trial is based on a built-in LSM9DS1 in Arduino Nano BLE 33 Sense. The sensor contains 3-axis accelerometer, 3-axis gyroscope, and 3-axis magnetometer. For this particular research, only the magnetometer is used.

To obtain the magnetometer data, the board is placed far from any magnetic material and rotated around it's axis multiple times. This way, one can expect the magnetometer measurement covers the whole direction and forms a sphere.

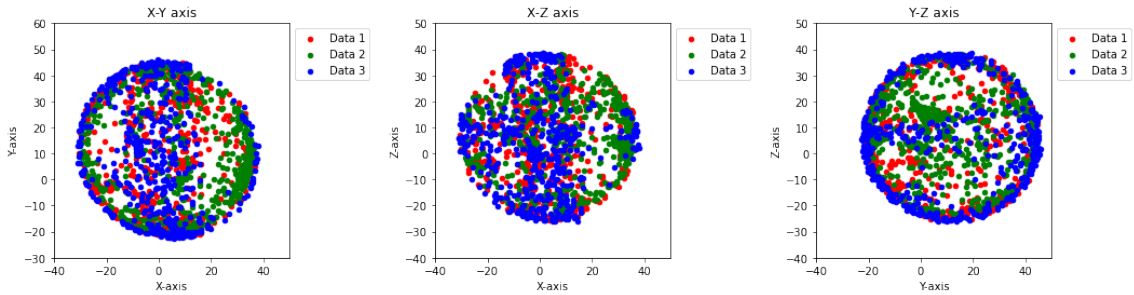

Figure 2: Magnetometer Measurement from Arduino Nano BLE 33 Sense

Figure 2 shows the magnetometer measurement from Ardunio Nano BLE 33 Sense. One can see that the sphere is not centered at the origin O. Thus, a calibration is required to find the offset of this measurement.

7.1 Sphere Fit

The calibration is based on the sphere fit problem as described in section 5. Levenberg-Marquardt Algorithm (as described in section 4) is used to find the optimum solution. There are three offset values obtained from three datasets, to check the consistency of the calibration result. The offset approximations are shown in table 1.

Figure 3 shows the offset approximation result in the magnetometer measurement plot. From this plot, it can be seen that the result is close to each other, thus the calibration method and value are acceptable.

Table 1: Sphere Fit Optimization Result

Data Index	**Offset**			**Radius**
	x	**y**	**z**	
1	3.964	12.048	7.337	33.676
2	3.284	11.676	7.103	33.196
3	3.349	11.768	6.239	35.016

Figure 4 shows the fitness parameter of the optimization. It can be seen that for data 1 the mean squared errors keeps decreasing at a slow rate. However, for data 2 and 3, the mean squared errors is constant after iteration 2.

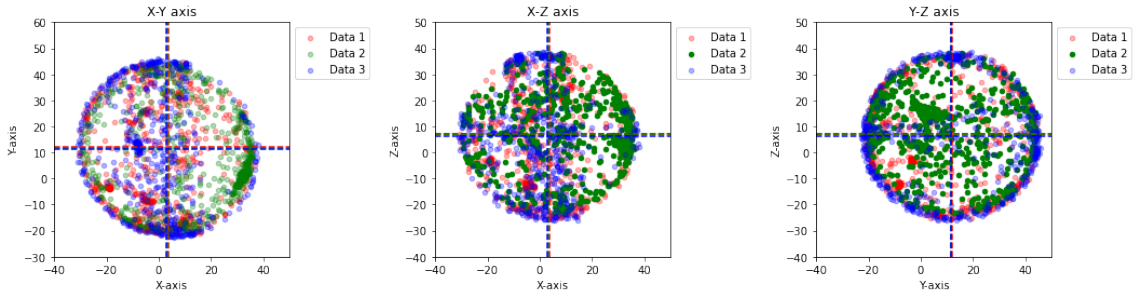

Figure 3: Offset Line Plot

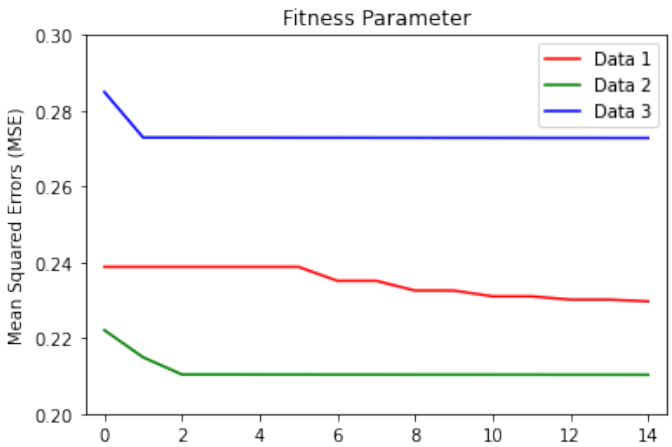

Figure 4: Sphere Fit - Fitness Parameter

Figure 5 shows the damping factor of the optimization. It can be seen that for all data, the damping factor does not converge to a value. Thus, one can conclude that the optimization process does not produce an optimum result.

One factor that contributes to the failure of achieving the optimum result is the sphere fit optimization limitation. This optimization assume that the data is a perfect sphere, having the same radius (or magnitude) in all direction. However, figure 6 shows that the measured magnetic field intensity is not a sphere as the value varies in a wide range.

In conclusion, the optimization process does not reach the optimum since the sphere fit method does not match the problem. One needs to evaluate the same data set on ellipsoid fit and compare

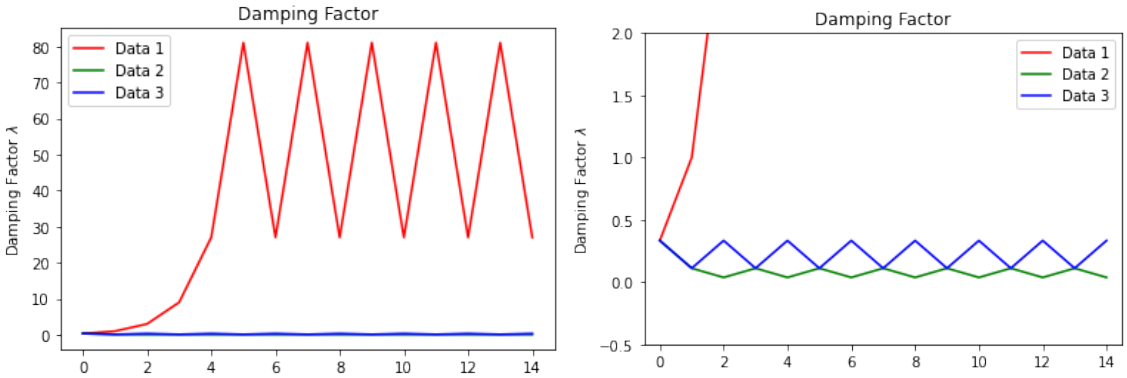

Figure 5: Sphere Fit - Damping Factor

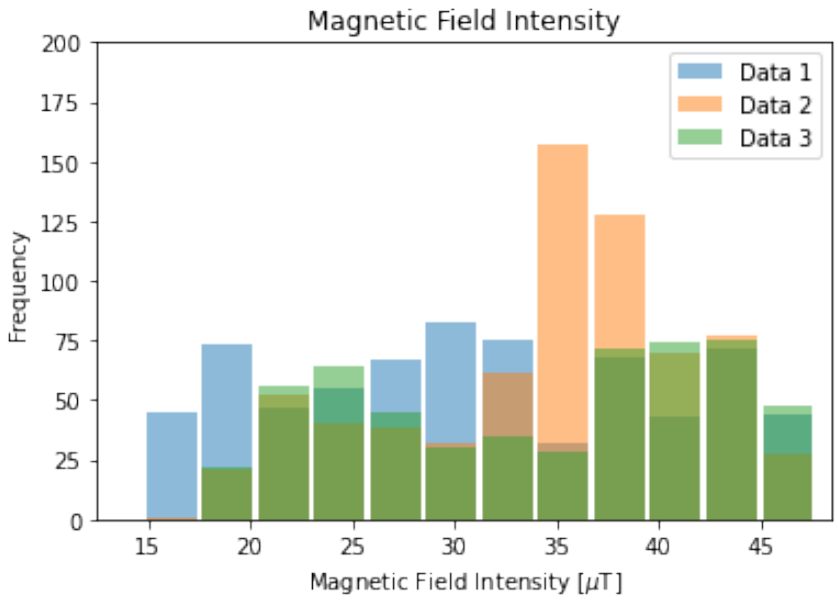

Figure 6: Histogram Plot of Magnetic Field Intensity

the result.

7.2 Ellipsoid Fit

The calibration is based on the sphere fit problem as described in section 6. Levenberg-Marquardt Algorithm (as described in section 4) is used to find the optimum solution. There are three offset values obtained from three datasets, to check the consistency of the calibration result. The offset approximations are shown in table 2.

Note that in the data index number 2, the x offset is far from the other data index in the same axis. This might be an indication of a calibration failure. One needs to check the fitness parameter and the damping ratio for the evaluation.

Figure 7 shows the mean squared errors for each data. It can be seen that at the end all optimization reach MSE below 0.1.

Table 2: Ellipsoid Fit Optimization Result

Data index	Offset		
	x	y	z
1	3.871	11.986	6.928
2	2.189	11.224	6.157
3	3.342	11.584	6.281

However, in figure 8, one can see that only data 1 and data 3 have the damping factor converge to a close to zero value. This means, the next iteration won't be far from the current point. Data 2 shows that the damping ratio is diverging, thus indicating a failure in the calibration.

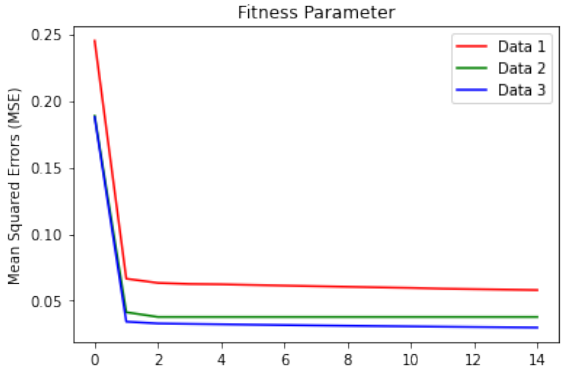

Figure 7: Ellipsoid Fit - Fitness Parameter

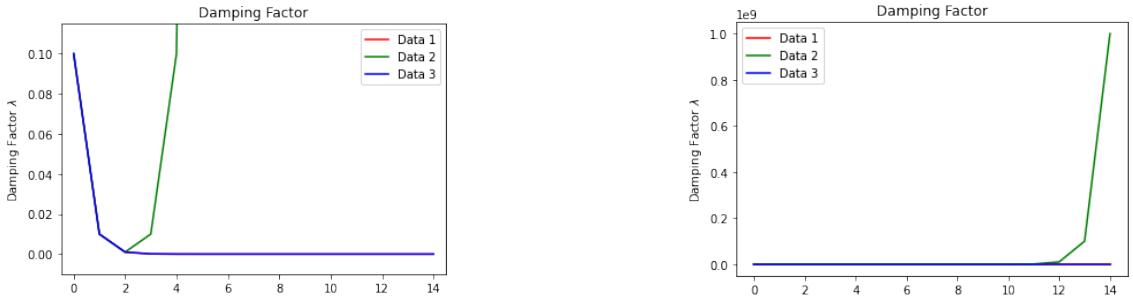

Figure 8: Ellipsoid Fit - Damping Factor

Figure 9 shows the magnetic intensity of uncalibrated and calibrated measurement. It can be seen that for data 2 the intensity has two value which has high frequency. This can be an indicator of the calibration failure. For data 1 and 3, most of the calibrated measurement is near 41 μT, which is the reference value for the magnetic intensity.

In conclusion, the damping factor can be an evaluation for the optimization performance along with the fitness parameter. This has been clarified by the result from data 2, where the x-offset is estimated far from the other data set and the damping factor is diverging. One needs to select a strict constraint for the fitness parameter since a diverging damping factor can have relatively small mean squared errors.

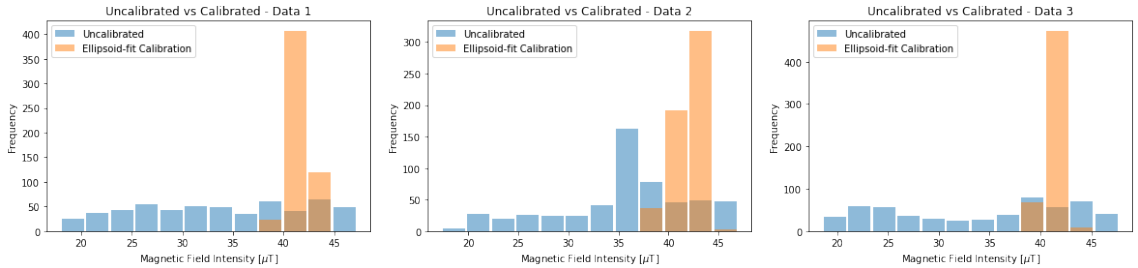

Figure 9: Histogram plot of ellipsoid fit result

8 Conclusion

In calibrating magnetometer, one can use sphere fit calibration or ellipsoid fit calibration. Sphere fit calibration requires 3 parameters from the bias vector and 1 parameter from the radius to be calculated. On the other hand, ellipsoid fit calibration requires 3 parameters from the bias vector and 6 parameters from the soft iron distortion matrix to be determined.

The calibration results shows that the ellipsoid fit produces lower mean squared errors compared to sphere fit algorithm. Also, the damping factor for ellipsoid fit converges for two data while non of the sphere fit damping factor converges. This means the ellipsoid fit is a better method for magnetometer calibration.

On evaluating the calibration performance, one can check on the fitness parameter and the damping factor. A good calibration produces a small fitness parameter, less than 0.1. The damping ratio of an acceptable calibration also converges to almost zero value.

More testing needs to be done to evaluate the performance of ellipsoid fit. A testing where the magnetometer is mounted on drone would provide a more realistic environment. An on flight testing also needs to be done to check the calibration performance.

9 Appendix

The partial derivative of the cost function with respect to one of bias vector, diagonal of soft iron matrix, and off-diagonal of soft iron matrix:

$$\frac{\partial f(\mathbf{x})}{\partial b_x} = -1 * \frac{A\frac{dA}{db_x} + B\frac{dB}{db_x} + C\frac{dC}{db_x}}{\sqrt{A^2+B^2+C^2}} = \frac{AK_{dx} + BK_{ox} + CK_{oy}}{\sqrt{A^2+B^2+C^2}}$$

$$\frac{\partial f(\mathbf{x})}{\partial K_{dx}} = -1 * \frac{A\frac{dA}{dK_{dx}} + B\frac{dB}{dK_{dx}} + C\frac{dC}{dK_{dx}}}{\sqrt{A^2+B^2+C^2}} = -1 * \frac{A(h_x - b_x)}{\sqrt{A^2+B^2+C^2}}$$

$$\frac{\partial f(\mathbf{x})}{\partial K_{ox}} - 1 * \frac{A\frac{dA}{dK_{ox}} + B\frac{dB}{dK_{ox}} + C\frac{dC}{dK_{ox}}}{\sqrt{A^2+B^2+C^2}} = -1 * \frac{A(h_y - b_y) + B(h_x - b_x)}{\sqrt{A^2+B^2+C^2}}$$